RETAILIQ: FULL-CYCLE E-COMMERCE CUSTOMER & SALES ANALYTICS

Final Project Review | Department of Computer Science and Engineering

TEAM MEMBERS

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Guide: Bonagani Prathusha, Faculty, CSE | Academic Year: 2026–27

Introduction & Problem Statement



Background

Online stores log thousands of orders, but raw transactions do not tell managers who their best customers are or how sales will move next.



Problem Statement

Retail data is noisy (duplicates, cancelled orders, missing IDs) and small retailers lack one tool to clean it, segment customers and forecast revenue.



Motivation / Need

Small gains in loyalty can raise profits considerably [1]; without insight, at-risk customers go unnoticed and revenue is lost.

THE DATA AT A GLANCE

541,909
raw transaction records

392,692
records after cleaning

4,338
customers segmented

£8.89M
revenue analysed (GBP)

Objectives & Scope

Objectives

1. Build a pipeline that cleans raw transaction data and derives revenue and time features.
2. Segment customers into interpretable behavioural groups using RFM analysis.
3. Tie each segment to a concrete retention or win-back action.
4. Project monthly revenue with an interpretable regression model.
5. Present the results in an interactive dashboard that starts from a drag-and-drop upload.
6. Add a natural-language assistant and a portable Docker deployment.

In Scope

- CSV/XLSX upload with schema checks
- Cleaning, revenue and time features
- RFM segmentation and retention actions
- 1–6 month linear revenue forecast
- Six-page dashboard, AI assistant, Docker

Out of Scope (future work)

- Churn and lifetime-value prediction
- Seasonal models (ARIMA, Prophet, LSTM)
- Forecast-accuracy metrics (MAE, RMSE, MAPE, R^2)
- Anomaly detection and multi-user cloud hosting

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Reichheld & Sasser, 1990	Customer-retention economics	Small gains in loyalty can raise profits considerably	Motivates retention but gives no analytics tool
2	Hughes, 1994	RFM scoring from direct marketing	Recency, frequency and monetary value describe customers simply	A concept; needs tooling to apply at scale
3	Christy et al., 2021	RFM ranking with K-Means and Fuzzy C-Means	RFM ranking is an effective approach to segmentation	Cluster count must be chosen; harder to explain
4	Chen, Sain & Guo, 2012	RFM with K-Means and decision trees on online retail data	Case study of RFM-based customer segmentation	Analysis only; no actions, forecast or dashboard

Research Gap

Prior work treats RFM and clustering as separate analyses. Small and medium retailers still lack one lightweight, explainable tool that cleans raw data, segments customers, links each segment to an action, forecasts revenue and answers questions in plain language.

Proposed System & Methodology

STEP 1

Ingest

Upload CSV/XLSX, normalise columns, reject invalid files.

STEP 2

Clean

Drop duplicates, cancellations and bad prices; add revenue features.

STEP 3

Segment

RFM quintile scores give six customer segments.

STEP 4

Forecast

Linear regression projects revenue for 1–6 months.

STEP 5

Deploy

Streamlit dashboard, AI assistant and Docker packaging.

Key Novelty / Contribution

One upload-first app for the full cycle: explainable RFM rules instead of black-box clusters, an action for every segment, and a data-grounded AI assistant with a rule-based fallback.

System Architecture

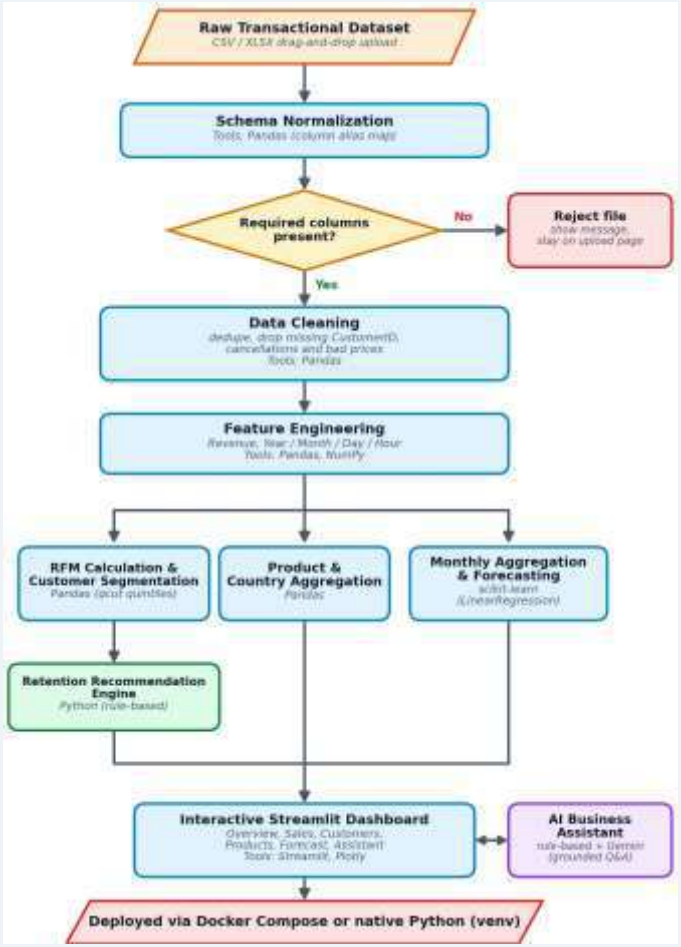


Fig. 1 Processing flowchart

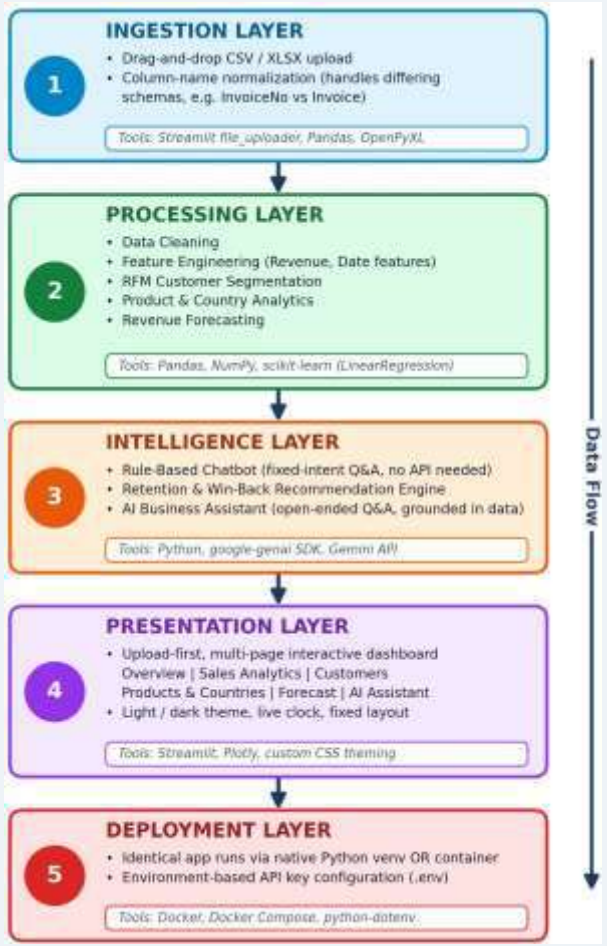


Fig. 2 Five-layer architecture

Components

1 Ingestion layer
CSV/XLSX upload, column normalisation and validation

2 Processing layer
Cleaning, features, RFM segments and forecast

3 Intelligence layer
Retention engine and AI assistant (rule-based + Gemini)

4 Presentation layer
Six-page Streamlit + Plotly dashboard, light/dark themes

5 Deployment layer
Native Python venv or Docker Compose

Hardware & Software Requirements

Hardware

- Standard laptop or desktop PC
- No GPU needed (light regression, API-based AI)
- Web browser to view the dashboard
- Internet only for the optional Gemini AI mode

Software & Tools

- OS: any that runs Python or Docker
- Python with a virtual environment (venv)
- Docker and Docker Compose
- Git version control (.env kept out of the repo)

Libraries / Frameworks

- Pandas, NumPy, OpenPyXL
- Scikit-learn (LinearRegression)
- Streamlit and Plotly
- Google GenAI SDK (Gemini Flash model)

Deployment / Platform: Local machine via native Python venv, or a Docker container (docker compose up --build) served on port 8501.

Dataset / Data Understanding

541,909

Raw records (rows)

8

Features (columns)

392,692

Cleaned records (72.5%)

4,338

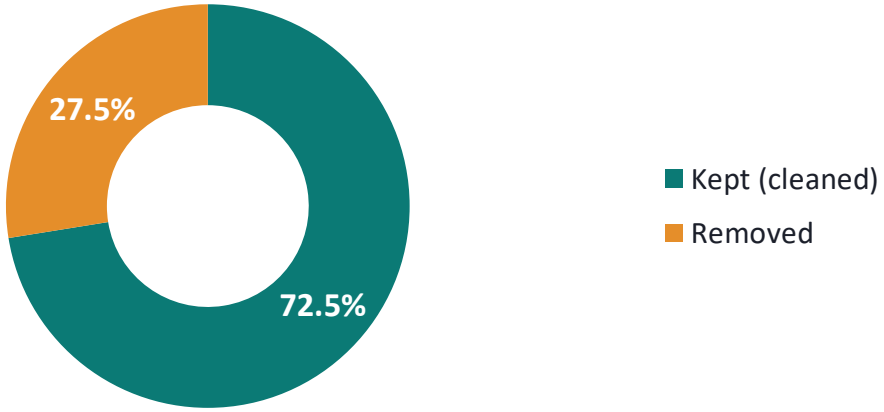
Unique customers

- Source: UCI Online Retail, UK online retailer (1 Dec 2010 – 9 Dec 2011)
- Features: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
- Quality: duplicates, missing CustomerID, cancellations and bad prices; 149,217 rows removed

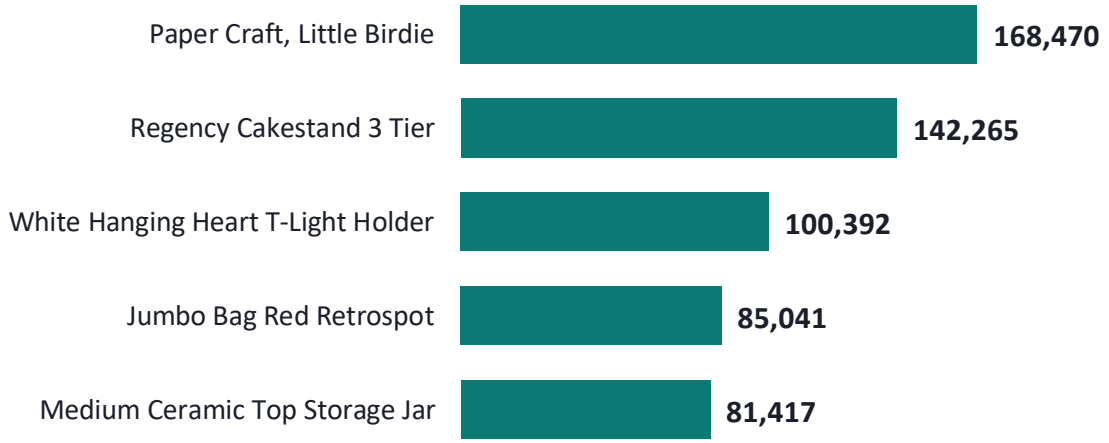
Attribute	Description
InvoiceNo	Transaction identifier
StockCode	Product identifier
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Transaction date and time
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country

Data Preprocessing & Exploratory Analysis

Records kept vs removed (541,909 raw)



Top 5 products by revenue (GBP)



Cleaning

Removed duplicates, missing CustomerID, cancellations and bad prices: 149,217 rows dropped.

Transformation

InvoiceDate to datetime; Revenue = Quantity × UnitPrice; Year, Month, Day, Hour extracted.

Feature Selection

Recency, Frequency and Monetary per customer, scored 1–5 (3 features).

Module-wise Description

1 Data Ingestion & Cleaning

Purpose: Validate, clean data and add revenue/time features

Input → Output: Raw CSV/XLSX → cleaned transactions

Developed by: Kolapally Vasavi

2 RFM Segmentation & Retention

Purpose: RFM scoring into six segments with retention actions

Input → Output: Cleaned data → RFM table + actions

Developed by: M. Haswanthi

3 Forecasting & Product Analytics

Purpose: Linear-trend forecast; product and country tables

Input → Output: Monthly revenue → 1–6 month forecast

Developed by: Kunchala Naga Chaitanya Sphoorthy

4 Dashboard & AI Assistant

Purpose: Six-page dashboard, AI assistant and Docker packaging

Input → Output: Analysis results → pages and answers

Developed by: All team members (integration)

Algorithms & Technologies Used

RFM Scoring

Why chosen
Easy for business users to read; no cluster count to choose

Key parameters
R, F, M scored 1–5 by quintiles (recency reversed); six rule-based segments

Linear Regression

Why chosen
Transparent trend baseline with a well-tested Scikit-learn version

Key parameters
$\hat{y}t = \beta 0 + \beta 1t$ on the month index t; horizon of 1–6 months

Gemini AI Assistant

Why chosen
Answers open questions from a compact summary of results, not the raw file

Key parameters
Gemini Flash via Google GenAI SDK; 3 retries with back-off, fallback model

Selection criteria: interpretability and simplicity. Quintile rules and a linear trend are easy to explain, and the rule-based mode keeps the assistant working without an API.

Implementation / Model Building

Input Features

Recency, Frequency, Monetary per customer, plus Year, Month, Day and Hour from InvoiceDate

Preprocessing Pipeline

Dedupe, drop missing CustomerID, cancellations and bad prices, parse dates, add Revenue

Training Data

Linear trend fitted on the full monthly revenue series; no train/test split or accuracy metrics yet

Configuration

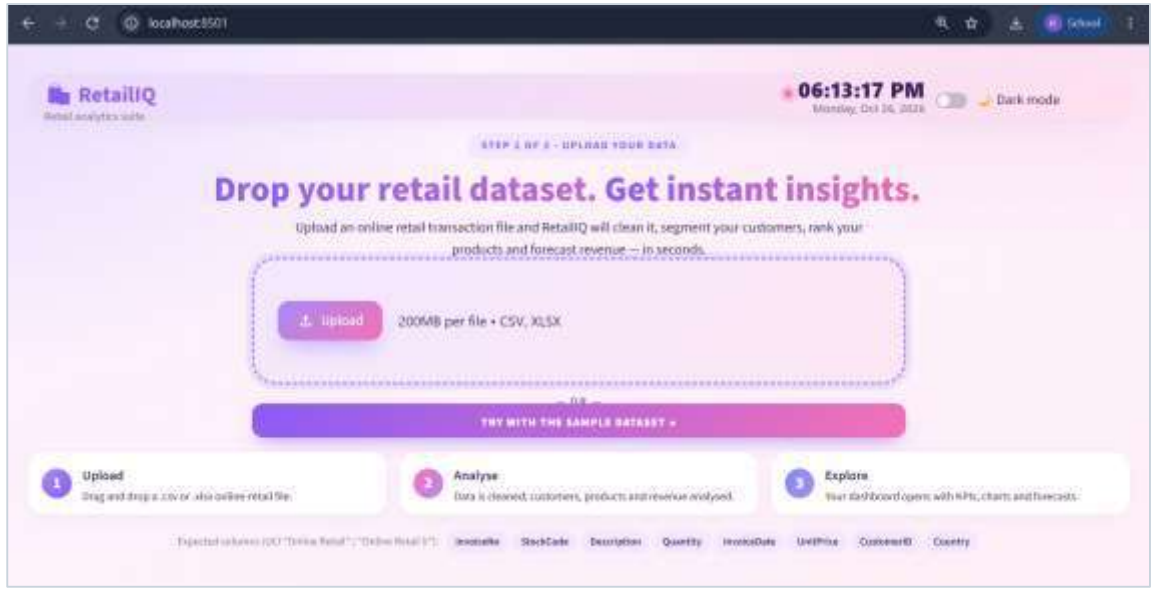
Horizon 1–6 months; date and country filters; cached results; API key from environment

Pseudocode (illustrative)

```
# RFM scoring (quintiles 1-5)
ref = last_date + 1 day
rfm = df.groupby('CustomerID').agg(
  Recency=days since last order,
  Frequency=nunique('InvoiceNo'),
  Monetary=sum('Revenue'))
rfm['R'] = qcut(Recency, 5, reversed)
rfm['F'] = qcut(Frequency, 5)
rfm['M'] = qcut(Monetary, 5)

# Revenue forecast: y = b0 + b1*t
model = LinearRegression().fit(t, revenue)
future = model.predict(next_months)
```

Working Prototype / Output Screens (1/4)





Upload page

- Drag-and-drop CSV or XLSX (200 MB per file)
- 'Try with the sample dataset' option
- Three-step guide: Upload, Analyse, Explore

Overview page

- Four KPI cards: revenue, orders, customers, average order value
- Monthly sales trend and sales-target ring
- Segment share, top products and module cards

Working Prototype / Output Screens (2/4)





Sales Analytics page

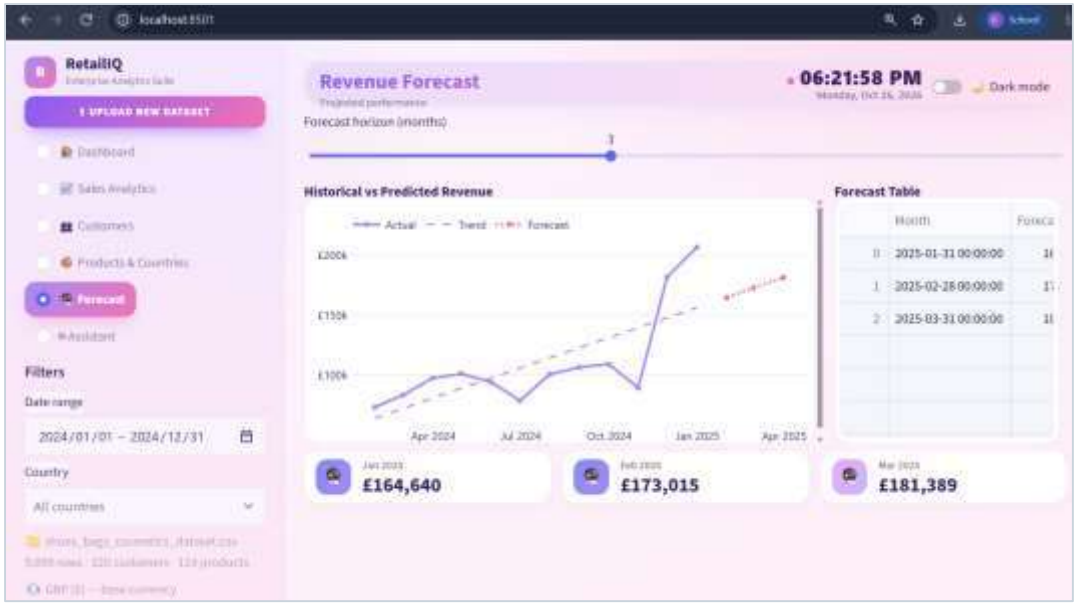
- Monthly revenue trend
- Orders by day of the week
- Orders by hour of the day

Customers page

- Customer KPIs and segment breakdown chart
- RFM customer table
- Retention & Win-Back tab with an action per segment

Working Prototype / Output Screens (3/4)





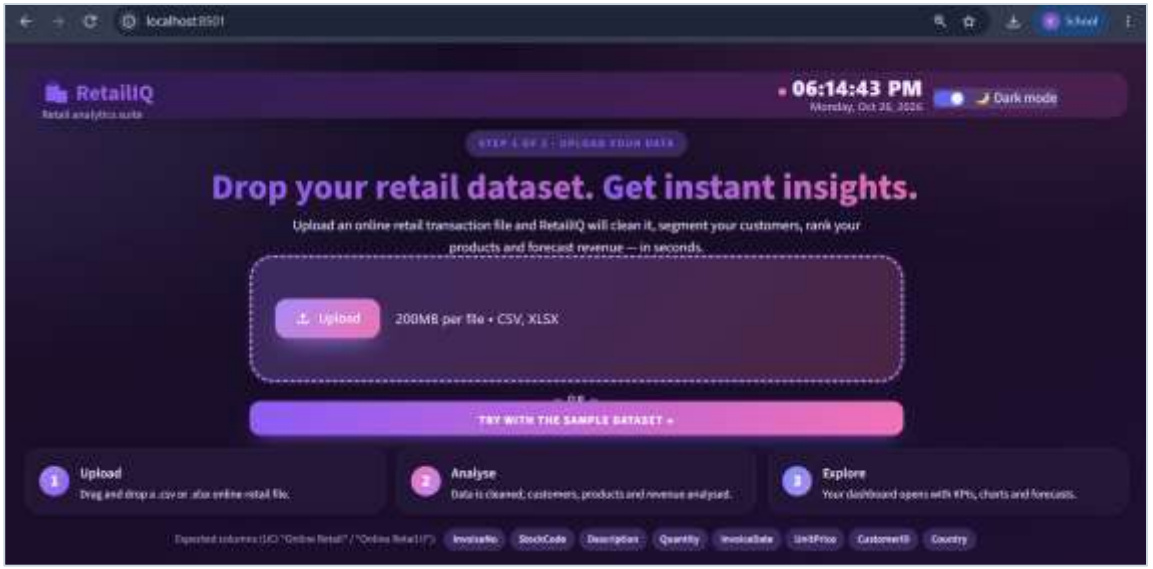
Products & Countries page

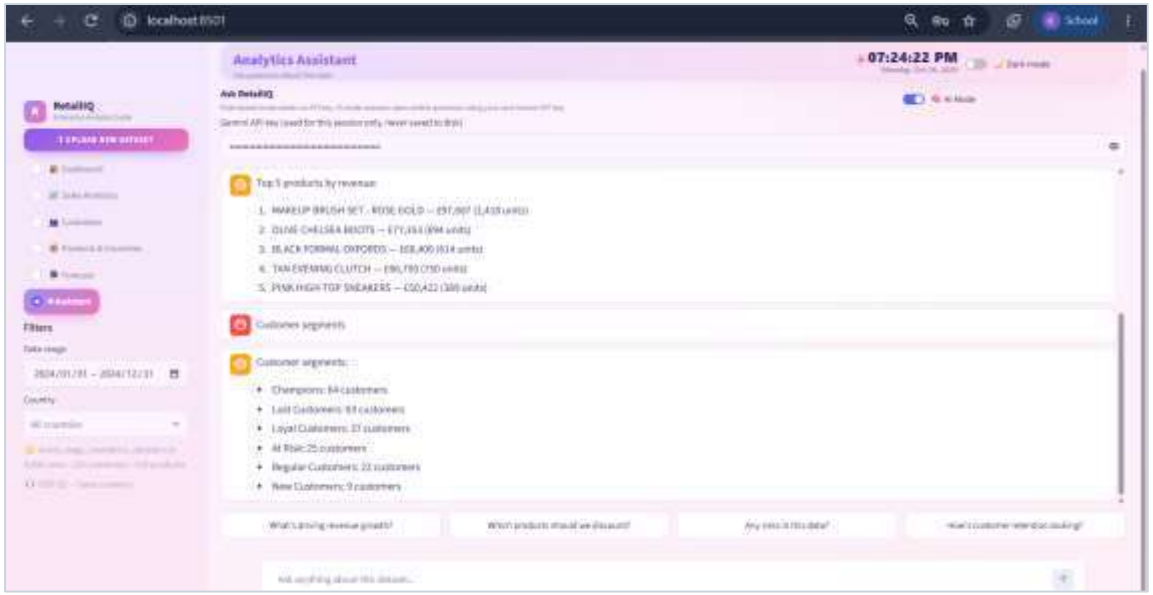
- Gallery of the top products
- Top-10 product table
- Country revenue chart and table

Forecast page

- Actual revenue, fitted trend and projection
- Forecast table and one card per month
- Horizon selectable from 1 to 6 months

Working Prototype / Output Screens (4/4)





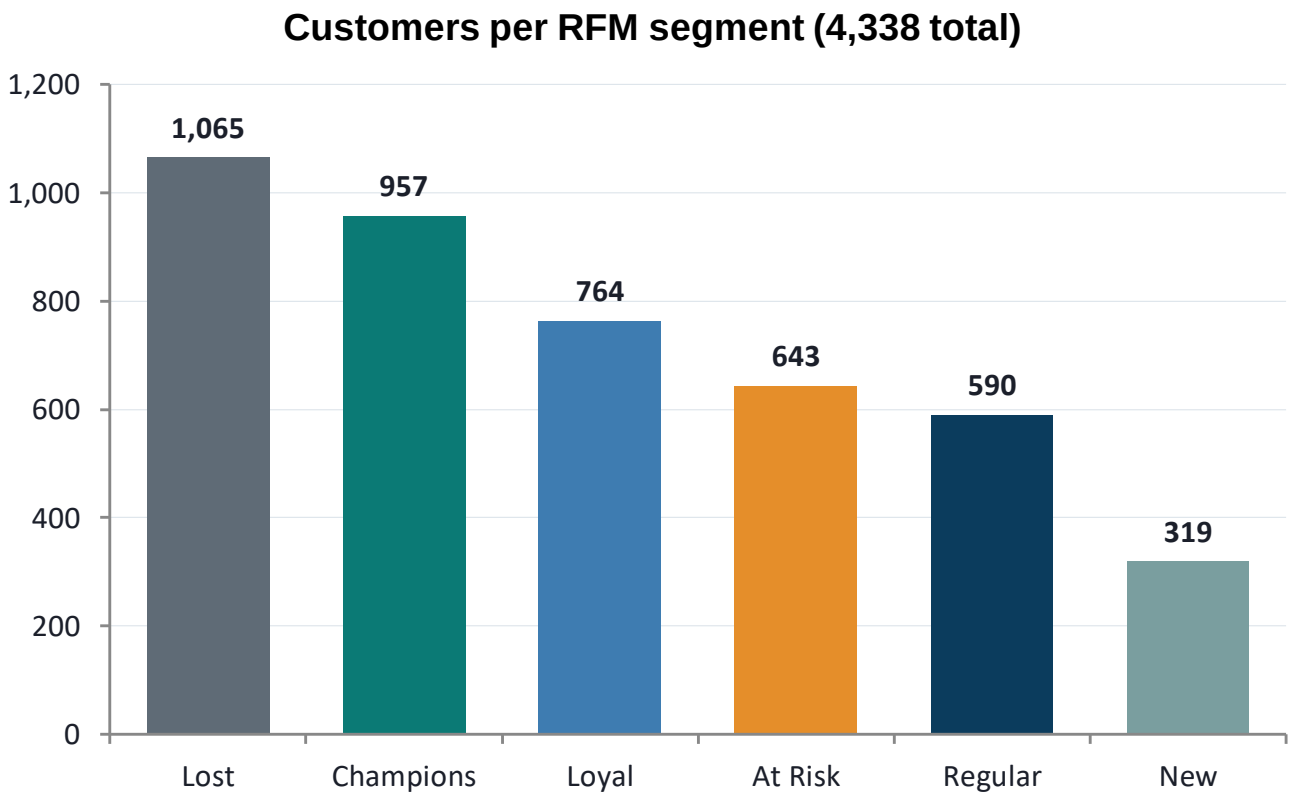
Dark theme

- Light/dark toggle in the top bar
- Live date and time, updated every second
- Fixed layout that fits one screen, no scrolling

AI Assistant page

- Chat interface with suggested questions
- Switch between rule-based and AI (Gemini) mode
- Answers grounded in the computed results

Results: Customer Segments (RFM)

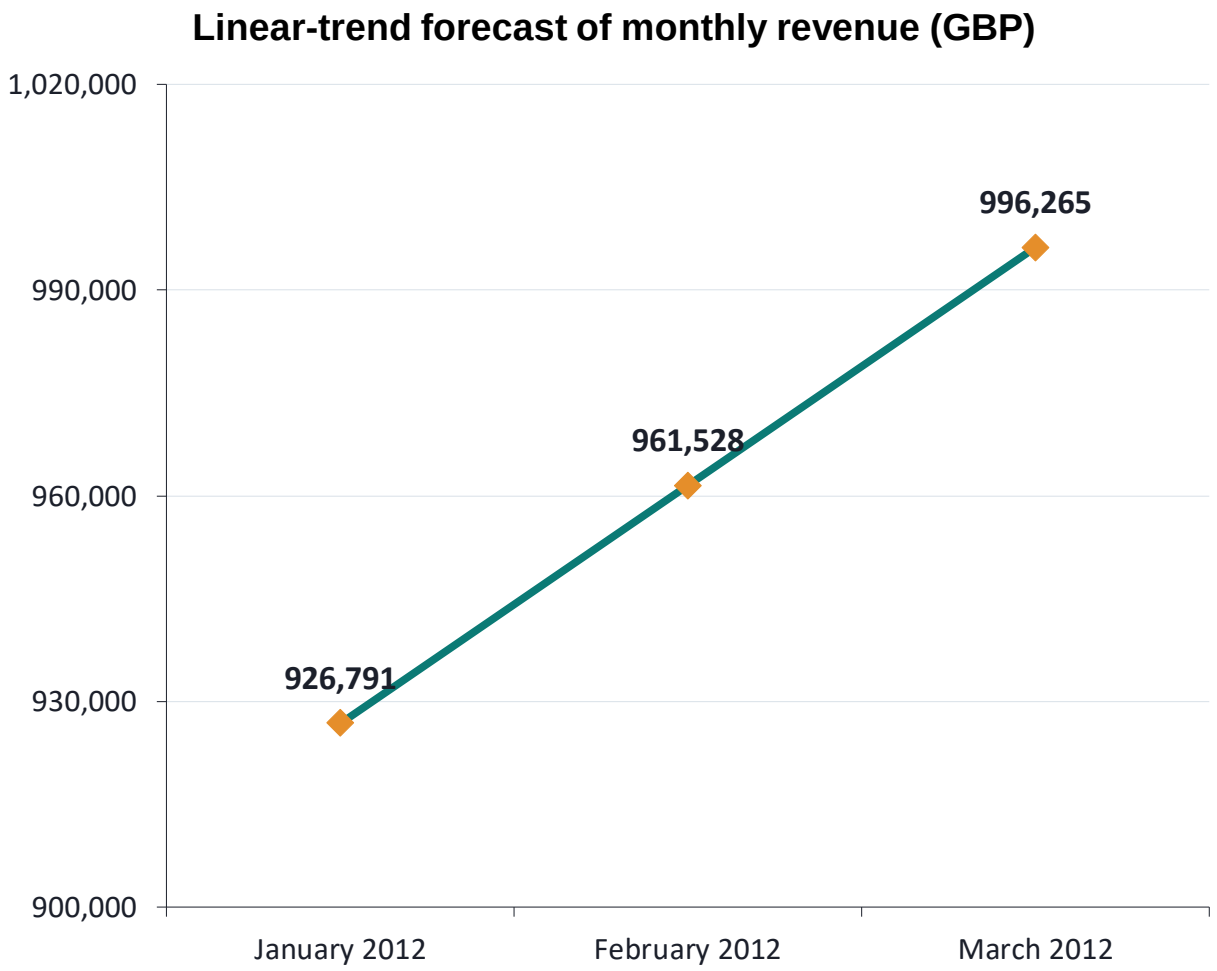


Segment	Customers	Share (%)
Lost Customers	1,065	24.6
Champions	957	22.1
Loyal Customers	764	17.6
At Risk	643	14.8
Regular Customers	590	13.6
New Customers	319	7.4
Total	4,338	100

Champions + Loyal ≈ 40% of customers

Source: UCI Online Retail dataset, 4,338 customers after cleaning. Lost customers are the largest group (24.6%).

Analysis & Key Findings



£34.7K

Revenue added each month by the fitted linear trend (β1 ≈ £34,737)

Interpretation: Lost customers are the largest segment (24.6%); Champions and Loyal are about 40%. Revenue is projected to rise steadily to March 2012.

Error analysis: No accuracy metrics yet (MAE, RMSE, MAPE, R^2); a straight trend ignores seasonality; the last month (to 9 Dec 2011) is incomplete.

Key drivers: The UK brings most revenue; the top product is Paper Craft, Little Birdie (£168,470).

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Open all six pages in light and dark themes	Pages load with no runtime errors	All pages opened without errors	Pass
TC-02	Upload a file missing required columns	Rejected with a message naming the columns	Rejected with a clear message	Pass
TC-03	Simulate AI service errors (503, rate limit)	Retry up to 3 times, then fall back	Retry logic handled the errors	Pass
TC-04	Clean the UCI Online Retail file	392,692 records, 4,338 customers	392,692 records, 4,338 customers	Pass



Functional Testing

Automated smoke tests opened all six pages in both themes without errors.



Validation Method

Upload check: invalid or incomplete files are rejected with a clear message.



Reliability Testing

Assistant retry logic tested with simulated service errors (3 retries, fallback model).

Challenges & Limitations

Challenges Faced

- Different column names across releases → alias map normalises the schema; invalid files rejected on upload
- Duplicates, cancellations, missing IDs, bad prices → six-step cleaning pipeline
- Overloaded hosted AI (HTTP 503, rate limits) → three retries with back-off, alternative model, rule-based fallback

Limitations

- Linear trend ignores seasonality; no forecast-accuracy metrics yet
- Data ends 9 Dec 2011, so the last month looks low; postage and manual items appear as products
- One country (UK) dominates; AI mode needs a Gemini key and internet; rule-based mode answers fixed questions only

Conclusion

1. RetailIQ is an upload-first Python and Streamlit application that cleans raw retail transactions and turns them into customer, product, sales and revenue insight.

2. On UCI Online Retail, 541,909 records were cleaned to 392,692, covering 4,338 customers, £8.89M revenue, 18,532 orders and an average order value of about £480.

3. All six objectives are met: cleaning pipeline, RFM segments with actions, revenue forecast, dashboard, AI assistant and Docker portability.



Final Outcome

4,338

customers segmented into six RFM groups
Objectives achieved: 6 of 6

Future Scope & Applications

Improve Performance
Try ARIMA, Prophet, XGBoost or LSTM; add MAE, RMSE, MAPE and R^2 accuracy checks

Expand Data & Features
More dataset formats, churn and lifetime-value prediction, unusual-sales detection

Smarter Assistant & Insight
Richer rule-based questions, explainable AI, cleaner product list, incomplete-month handling

Scale & Recommend
Cloud hosting for many users, plus better product and customer recommendations

Applications

- Retention and win-back campaigns by segment
- Monthly revenue planning and trend tracking
- Product and country performance reviews
- Adopters: small and medium online retailers and wholesalers

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030785	Kolapally Vasavi	Part 1 – Data & analytics pipeline: upload validation, schema normalisation, cleaning, feature engineering, RFM scoring and segments, retention playbook; notebooks;
2420030510	Kunchala Naga Chaitanya Sphoorthy	Part 2 – Analytics, forecast & assistant: product/country analysis, currency, revenue forecast and backtest, rule-based chatbot, Gemini assistant with retry/fallback, testing;
2420030143	M. Haswanthi	Part 3 – Dashboard & deployment: Streamlit pages and upload flow, light/dark theme, live clock, fixed layout, Docker/Compose.

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Project Links

Dataset: https://archive.ics.uci.edu/dataset/352/online+retail

Demo: docker compose up --build → localhost:8501

Run (native): streamlit run dashboard/app.py

THANK YOU

Questions & Discussion

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